Durgesh Dutt Sinha

AIML Engineer & Full-Stack Developer

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EXECUTIVE SUMMARY

Forward-thinking **AIML Engineer & Full-Stack Architect** pursuing an MCA in Artificial Intelligence & Machine Learning at Sri Balaji University, Pune (2025–2027). Selected as an **UNLOX® AI Program Fellow** and **Be10x AI Cohort Member**. Demonstrates proven delivery of production-ready systems including real-time computer vision pose estimation engines (<50ms inference), enterprise ATS resume evaluation platforms with Google XYZ bullet transformers, and unsupervised retail customer intelligence suites.

TECHNICAL COMPETENCIES

AI & Machine Learning: Autonomous AI Agents, Prompt Engineering, Google MediaPipe Computer Vision, Scikit-Learn (K-Means, GMM, DBSCAN), PyTorch, CNNs, Whisper STT.

Full-Stack & Cloud: Next.js 16/15 (App Router), React 19/18, TypeScript 5.x, Drizzle ORM, PostgreSQL (Neon), Tailwind CSS v4, WebSockets, FastAPI, Docker.

Architecture & DevOps: Cloudflare Pages Edge CDN, AWS Cloud ML Pipelines, GitHub Actions CI/CD, Dual-Mode Storage Resiliency, Web Workers concurrency.

Practices & Tooling: System Architecture Design, Clean Code, Test-Driven Development, WCAG 2.1 AA Accessibility, Git / GitHub, Turbopack, Vite.

PRODUCTION ENGINEERING PROJECTS

RoleRadar — AI Career Intelligence & ATS Resume Engineering Platform — July 2026 – Present

Tech Stack: Next.js 16.2 • React 19 • TypeScript 5.9 • Drizzle ORM • PostgreSQL (Neon) • Tailwind CSS v4 • pdf-parse & mammoth

- Evaluates candidate resumes across 16 modern tech roles with an automated 8-point ATS compatibility scanner, dynamic Google XYZ bullet optimizer, and sub-180ms scoring.
- Built robust document parsing pipeline (.pdf/.docx), keyword normalizer against 16-role benchmark taxonomies, and zero-crash dual-mode storage (Neon PostgreSQL with in-memory fallback).

FitTrack AI Fitness Platform — Edge Computer Vision & Real-Time Biomechanics — August 2026 – Present

Tech Stack: TypeScript • Google MediaPipe Pose • Web Workers API • WebSockets • Canvas 2D / WebGL • Tailwind CSS

- Architected 9-page multi-modal fitness architecture utilizing Google MediaPipe 33-point skeletal landmark detection and trigonometric joint-angle vector kinematics under 50ms latency.
- Piped 720p video frames into dedicated Web Worker via transferable ImageBitmap for background inference, keeping the main UI thread lag-free during 60fps biomechanical analysis.

MarketMatch-AI — Enterprise E-Commerce RFM Customer Intelligence Studio — July 2026 – Present

Tech Stack: Python 3.11 • Scikit-Learn • Streamlit • Plotly 3D • Pandas & NumPy • Joblib

- Ingested 4,000+ retail transaction records; computed RFM metrics and engineered customer segmentation cohorts using K-Means ($K=5$), Gaussian Mixture Models (GMM), and DBSCAN anomaly detection.
- Applied Log1p scaling and StandardScaler to normalize skewed monetary distributions; produced interactive 3D visualizations identifying high-value lookalike customer segments.

FELLOWSHIPS & ENGINEERING EXPERIENCE

AI Program Fellow — UNLOX® — Jun 2026 – Present | Remote

- Engineered end-to-end AI deployment pipelines with automated containerization; constructed and benchmarked autonomous agent workflows for enterprise industry use-cases under technical mentorship.

AI Cohort Member — Be10x — May 2026 – Present | India

- Developed multi-step LLM prompt chaining architectures to automate repetitive tasks; built full-stack AI concept prototypes from initial system design to live client demonstrations.

EDUCATION

Master of Computer Applications (AIML) — Jul 2025 – 2027 (Expected)
Sri Balaji University, Pune
Specialization: Artificial Intelligence & Machine Learning

Bachelor of Computer Applications — Aug 2022 – Apr 2025
Sri Balaji University, Pune
Focus: AI, System Architecture, Prompt Engineering

CREDENTIALS & CERTIFICATIONS

- **AWS Educate Emerging Talent Community Member**
  Amazon Web Services (AWS)
- **AWS Educate Machine Learning Foundations**
  Amazon Web Services (AWS)